Database Schema Design

Practicum I

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Part B:- Normalized Database Design

Data Sources

The data comes from two sources:

1. **CSV file** (`restaurant-visits-209874.csv`) — one row per restaurant visit with columns: `RestaurantID`, `ServerEmpID`, `ServerName`, `VisitDate`, `VisitTime`, `MealType`, `PartySize`, `WaitTime`, `CustomerName`, `CustomerPhone`, `CustomerEmail`, `LoyaltyMember`, `FoodBill`, `TipAmount`, `FoodDiscountPercentage`, `PaymentMethod`, `orderedAlcohol`, `AlcoholBill`.
2. **SQLite database** (`restaurants-db-02102026.sqlitedb`) — two tables:
   - `restaurants(rID, RestaurantName, City, State, hasTableService)`
   - `servers(EmpID, ServerName, StartDateHired, EndDateHired, HourlyRate, BirthDate, SSN, rID)`

Question B(1) — Functional Dependencies

Let R be the universal relation combining all attributes from both sources. The attributes are:

R = { RestaurantID, RestaurantName, City, State, hasTableService, ServerEmpID, ServerName, StartDateHired, EndDateHired, HourlyRate, BirthDate, SSN, VisitDate, VisitTime, MealType, PartySize, WaitTime, CustomerName, CustomerPhone, CustomerEmail, LoyaltyMember, FoodBill, TipAmount, FoodDiscountPercentage, PaymentMethod, orderedAlcohol, AlcoholBill }

We denote a visit by the composite key $(RestaurantID, ServerEmpID, VisitDate, VisitTime)$ or, in the normalized form, a surrogate `VisitID`.

Restaurant FDs:

$$RestaurantID \rightarrow RestaurantName, City, State, hasTableService$$

Server FDs:

$$ServerEmpID \rightarrow ServerName, StartDateHired, EndDateHired, HourlyRate, BirthDate, SSN, RestaurantID$$

$$SSN \rightarrow ServerEmpID, ServerName, BirthDate$$

Customer FDs:

$$CustomerPhone \rightarrow CustomerName, CustomerEmail, LoyaltyMember$$

$$CustomerEmail \rightarrow CustomerName, CustomerPhone, LoyaltyMember$$

Visit FDs (with surrogate `VisitID` or natural key `(RestaurantID, VisitDate, VisitTime)`):

$$VisitID \rightarrow RestaurantID, ServerEmpID, VisitDate, VisitTime, MealType, PartySize, WaitTime, CustomerID, FoodB$$

Payment method is a categorical attribute; no further FDs decompose it.

MealType is a categorical attribute determined solely by itself (lookup value).

Question B(2) — 3NF Decomposition

Starting from R, we identify and remove all partial and transitive dependencies.

Step 1 — Remove restaurant transitive dependency: Extract `restaurants(restaurant_id PK, restaurant_name, city, state, has_table_service)`.

Step 2 — Remove server transitive dependency: Extract `servers(emp_id PK, server_name, start_date_hired, end_date_hired, hourly_rate, birth_date, ssn, restaurant_id FK)`.

Step 3 — Remove customer transitive dependency: Extract `customers(customer_id PK, customer_name, customer_phone, customer_email, loyalty_member)`. Customers are identified by a surrogate key; phone and email are candidate keys when not null.

Step 4 — Lookup tables for categoricals: Extract `meal_types(meal_type_id PK, meal_type_name)` and `payment_methods(payment_method_id PK, payment_method_name)`.

Step 5 — Remaining visit attributes form the visits relation: `visits(visit_id PK, restaurant_id FK, emp_id FK, customer_id FK, visit_date, visit_time, meal_type_id FK, party_size, wait_time, food_bill, tip_amount, food_discount_pct, payment_method_id FK, ordered_alcohol, alcohol_bill)`.

All resulting relations are in **3NF**: every non-key attribute is fully functionally dependent on the entire primary key, and there are no transitive dependencies among non-key attributes.

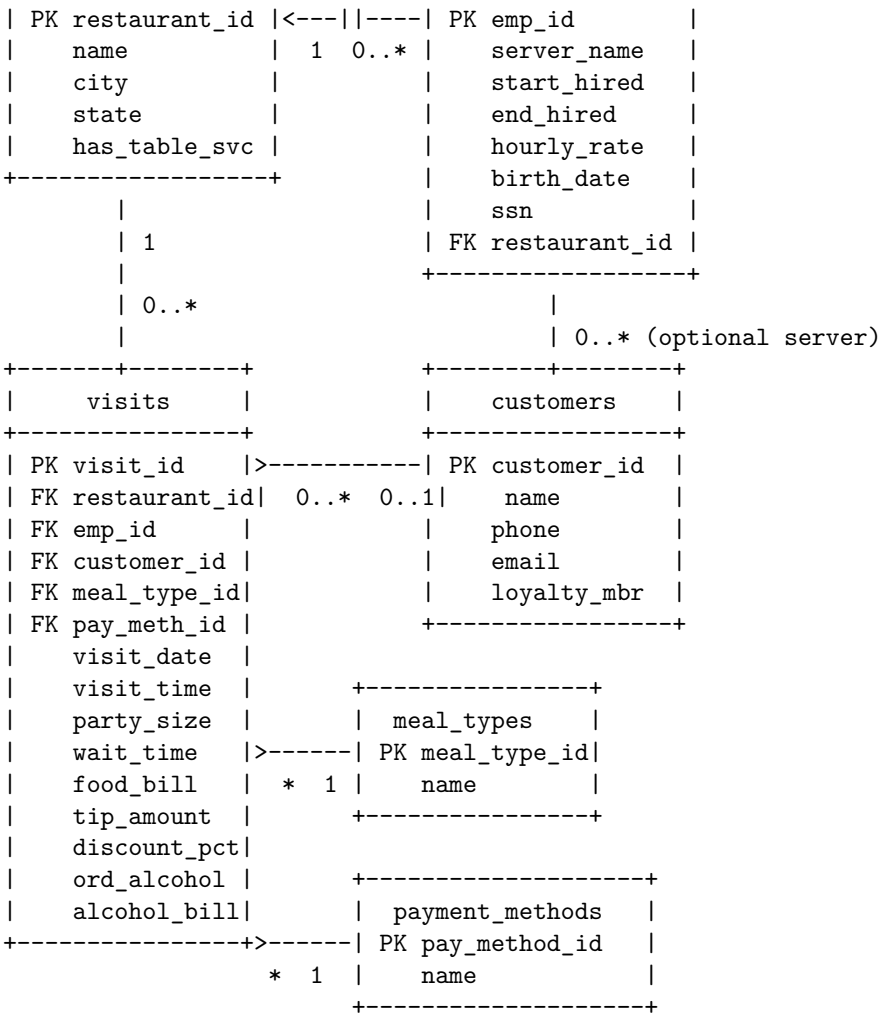
Final relations:

Relation	Primary Key	Foreign Keys
restaurants	restaurant_id	—
servers	emp_id	restaurant_id
customers	customer_id	—
meal_types	meal_type_id	—
payment_methods	payment_method_id	—
visits	visit_id	restaurant_id, emp_id, customer_id, meal_type_id, payment_method_id

Question B.3 — ERD (IE / Crow's Foot Notation)

The ERD is described below in textual Crow's Foot notation. Each entity is listed with its attributes, and relationships show cardinality.





Cardinality Notes:

- One `restaurant` has zero or many `visits`; each `visit` belongs to exactly one `restaurant`.
- One `restaurant` employs zero or many `servers`; each `server` belongs to exactly one `restaurant`.
- Each `visit` is optionally served by one `server` (server may be unknown); a `server` may have zero or many `visits`.
- Each `visit` is optionally associated with one `customer` (anonymous visits allowed); a `customer` may have zero or many `visits`.
- Each `visit` references exactly one `meal_type` and exactly one `payment_method`.

Question B(4) — ERD Embedded

The ERD is embedded above as an ASCII Crow's Foot diagram. For a graphical version, the same schema was modeled in LucidChart and is described by the textual representation in Section B.3.

All relations satisfy 3NF: no partial dependencies exist (all non-key attributes depend on the full primary key), and no transitive dependencies exist (non-key attributes do not determine other non-key attributes within the same relation).